

Database Architecture Demonstration Packet

VSEPR-SIM Day 40C Report Suite — Reports 1–10

Typed Scientific Registry for Atomistic, Material, Compound, and Precursor Records

Contents

§1 Database Architecture Demonstration Reports

The database layer is treated as a typed scientific registry rather than a flat storage table. Each input is modeled as

$$X_i = (I_i, T_i, D_i, P_i, S_i, H_i), \quad (1)$$

where I_i is the identity block, T_i is the type block, D_i is the dimension descriptor, P_i is the scientific payload, S_i is the deterministic seed block, and H_i is the hash and provenance block.

The architecture is designed to host:

- elemental records,
- compound and material records,
- precursor staging objects,
- mixed-resolution simulation inputs,
- deterministic reconstruction using seed and hash provenance.

The flow is:

Input Layer $\longrightarrow$ Canonical Registry $\longrightarrow$ Projection / Resolution Layer $\longrightarrow$ Validation / Hash / Report

§2 Report 1: Input Dimension Demonstration

§2.1 Goal

This report demonstrates how simulation and database inputs are described in terms of dimensionality, resolution level, physical scale, storage type, and projection target. The intent is to ensure that every input object has a known shape before entering either the atomistic engine, the bead engine, or the formation engine.

§2.2 General Input Form

For a generic scientific record X_i , the six blocks of Eq. (??) expand as follows:

Block	Description
I_i	Identity block (name, symbol, canonical ID)
T_i	Type block (element, compound, material, precursor, ...)
D_i	Dimension block (see below)
P_i	Payload block (physical / chemical / nuclear data)
S_i	Seed / deterministic initialization block
H_i	Hash / provenance block

§2.3 Dimension Block

A dimension descriptor is written as

$$D_i = (n_{\text{state}}, n_{\text{geom}}, n_{\text{chem}}, n_{\text{nuc}}, n_{\text{aux}}), \quad (2)$$

allowing a single registry to host element cores, isotope packets, atomistic objects, bead objects, materials, compounds, and staged precursors.

Component	Meaning
n_{state}	Number of runtime state variables
n_{geom}	Spatial / geometric dimensions
n_{chem}	Chemical descriptor count
n_{nuc}	Nuclear descriptor count
n_{aux}	Metadata / diagnostics / tags

§2.4 Input Dimension Table

Input Class	Repr.	n_{geom}	n_{chem}	n_{nuc}	Typical Payload
Element core	scalar record	0	4–12	0–4	name, symbol, Z
Isotope record	scalar record	0	2–8	4–12	decay, half-life
Atomistic object	vector state	3	6–20	0–8	$\mathbf{r}, \mathbf{v}, q, \sigma_{\text{chem}}, \sigma_{\text{nuc}}$
Bead object	reduced state	3	4–16	0–6	$R, V, Q, \Theta, \beta, \chi$
Compound record	graph / set	0–3	8–64	0–12	stoichiometry, class
Material preset	fielded record	0–3	10–48	0–8	density, phase, modulus
Precursor object	staged object	0–3	6–32	0–8	reactive flags, seed geometry

§2.5 Demonstration Inputs

Example A: Hydrogen element core.

$$X_H^{\text{core}} = (Z=1, \text{name=Hydrogen, symbol=H, class=nonmetal}).$$

Example B: Plutonium atomistic object.

$$X_{\text{Pu}}^{(a)} = (\mathbf{r}, \mathbf{v}, q, \sigma_{\text{chem}}, \sigma_{\text{nuc}}),$$

with

$$\sigma_{\text{nuc}} = (\lambda, E_{\text{dec}}, N_{\text{emit}}, \tau, \text{mode}).$$

Example C: Cyclopentadienyl precursor bead.

$$X_{\text{Cp}}^{(b)} = (R, V, Q, \Theta, \beta, \chi, \eta).$$

This demonstrates that the database accepts multiple dimensional classes without pretending they are all the same object.

§3 Report 2: Data Type Demonstration

§3.1 Goal

This report demonstrates the internal data typing rules used by the architecture. Every field is assigned a *semantic type* and a *storage type*.

§3.2 Typed Record Form

A field descriptor is:

$$F_{ij} = (\text{name, semantic_type, storage_type, units, validity}). \quad (3)$$

§3.3 Semantic Type Classes

Semantic Type	Meaning	Example
identity	Immutable object identifier	$Z=94$, Pu
categorical	Discrete class label	actinide, ligand, precursor
scalar	Single real/integer value	mass, radius, charge
vector	Ordered numeric list	position, velocity
tensor	Matrix-like or directional data	inertia, polarizability
set	Unordered discrete collection	oxidation states
graph	Connectivity relation	bonding map
hash	Content signature	SHA-256 / custom 562
pathref	Path to canonical shard	/elements/094_Pu/core.json

§3.4 Storage Type Classes

Storage Type	Use
u32	Compact indices, element IDs
u64	Extended object IDs
f32	Fast approximate simulation values
f64	Scientific numeric values
str	Names, symbols, tags
bool	Flags, response switches
vec<T>	Ordered arrays
set<T>	Non-duplicate category collections
map<K,V>	Keyed lookup data
hash256	Record integrity (SHA-256)
hash562	Deep provenance digest (custom)

§3.5 Demo Schema

Field	Semantic	Storage	Units	Validity
atomic_number	identity	u32	—	immutable
half_life_s	scalar	f64	s	isotope-only
oxidation_states	set	vec<i32>	—	context-sensitive
bead_orientation	tensor	vec<f64>	rad	bead projection only

§4 Report 3: Material Demonstration Report

§4.1 Goal

This report demonstrates how engineering and chemistry materials are stored as typed overlays rather than collapsed into element records.

A material is not an element.

§4.2 Material Record Form

$$M_k = (I_k, C_k, \rho_k, \phi_k, \Pi_k, \Gamma_k, H_k), \quad (4)$$

where:

Block	Description
I_k	Material identity
C_k	Composition (stoichiometric or mass-fraction)
ρ_k	Density / bulk density
ϕ_k	Phase state
Π_k	Engineering property vector
Γ_k	Hazard / environment descriptor
H_k	Provenance hash block

§4.3 Example Materials

A. Copper.

$$M_{\text{Cu}} = (\text{Copper}, \{\text{Cu:1.0}\}, \rho, \text{solid}, \Pi, \Gamma, H).$$

B. Steel preset (A36).

$$M_{\text{Steel-A36}} = (\text{A36}, \{\text{Fe, C, Mn, ...}\}, \rho, \text{solid}, E, \sigma_y, \nu, \dots).$$

C. Plutonium oxide-bearing residue.

$$M_{\text{PuResidue}} = (\text{mixed residue}, \{\text{Pu, O, C, H, trace}\}, \rho, \phi, \Pi, \Gamma, H).$$

§4.4 Demo Material Table

Material	Class	Phase	Main Use
Copper	elemental material	solid	conductor, engineering
Iron	elemental material	solid	base metal
Uranium dioxide	compound material	solid	nuclear fuel model
Thorium oxalate	precursor / compound	solid	slurry / precipitation / staging
Hexene	organic precursor	liquid	bead-scale precursor
Cyclopentadienyl prec.	organometallic ligand	molecular	hybrid architecture tests

§5 Report 4: Compound Demonstration Report

§5.1 Goal

This report demonstrates how compounds are encoded as distinct database objects with stoichiometric, structural, and resolution-aware metadata.

§5.2 Compound Record Form

$$C_j = (S_j, G_j, B_j, P_j, R_j, H_j), \quad (5)$$

where:

Block	Description
S_j	Stoichiometry
G_j	Geometry / graph / arrangement hints
B_j	Bond or interaction classification
P_j	Phase / process metadata
R_j	Resolution projections
H_j	Compound hash block

§5.3 Demonstration Compounds

A. Water.

$$C_{\text{H}_2\text{O}} = (\{\text{H}:2, \text{O}:1\}, \text{ bent, polar covalent, molecular, } R, H).$$

B. Thorium oxalate.

$$C_{\text{Th-ox}} = (\{\text{Th}, \text{C}, \text{O}, \text{H}\}, \text{ coordination / precipitation, ionic-covalent hybrid, precursor solid, } R, H).$$

C. Uranium dioxide.

$$C_{\text{UO}_2} = (\{\text{U}:1, \text{O}:2\}, \text{ crystal lattice, ionic / lattice, ceramic solid, } R, H).$$

D. Hexene.

$$C_{\text{C}_6\text{H}_{12}} = (\{\text{C}:6, \text{H}:12\}, \text{ chain / isomer family, covalent organic, molecular liquid, } R, H).$$

§5.4 Compound Demo Payload

Listing 1: Hexene compound registry entry

```
{
  "compound_id":  "cmp_hexene_001",
  "name":         "hexene",
  "formula":      "C6H12",
  "class":        "organic_precursor",
  "stoichiometry": {"C": 6, "H": 12},
  "structure_family": "alkene",
  "projection_modes": ["atomistic", "bead"],
  "default_seed_class": "chain_seed_v1",
  "hash256": "....",
  "hash562": "...."
}
```

§6 Report 5: Precursor Demonstration Report

§6.1 Goal

This report demonstrates how precursors are treated as staged formation objects rather than fully committed final structures. A precursor is not just a compound. It is a reaction-positioned entity.

§6.2 Precursor Record Form

$$P_m = (I_m, \Psi_m, \Omega_m, \Lambda_m, T_m, H_m), \quad (6)$$

where:

Block	Description
I_m	Precursor identity
Ψ_m	Chemical state
Ω_m	Environment readiness
Λ_m	Transformation / reaction flags
T_m	Target formation class
H_m	Provenance block

§6.3 Example Precursors

Precursor	Class	Target Formation
Hexene bead prec.	organic chain precursor	bead-tested hydrocarbon family
Cyclopentadienyl ligand prec.	organometallic precursor	mixed bead/atomistic coordination
Thorium oxalate slurry prec.	precipitation precursor	oxide or residue evolution
Dissolved nitrate-metal complex	process precursor	sludge / salt / oxide outcome
Pipe-scale dissolved solids	process-field precursor	materials + deposition model

§6.4 Hybrid Precursor Demo

A mixed-resolution precursor packet:

$$P_{\text{hybrid}} = (X_{\text{Pu}}^{(a)}, X_{\text{Cp}}^{(b)}, X_{\text{hexene}}^{(b)}, E, T, H), \quad (7)$$

where $X_{\text{Pu}}^{(a)}$ is atomistic plutonium, $X_{\text{Cp}}^{(b)}$ is bead-scale cyclopentadienyl precursor, $X_{\text{hexene}}^{(b)}$ is bead-scale hexene precursor, E is the environment state, and T is the transformation intent.

§7 Report 6: Seed and Hash Demonstration Report

§7.1 Goal

This report demonstrates deterministic reconstruction and provenance control. Every major input carries both a seed for reproducible generation and a hash for identity and integrity verification.

§7.2 Seed Block

$$S_i = (\text{seed}_{32}, \text{seed}_{64}, \text{seed}_{\text{geom}}, \text{seed}_{\text{chem}}), \quad (8)$$

where seeds may control:

- randomized cloud generation,
- bead placement,
- lattice jitter,
- initial velocity assignment,
- synthetic precursor arrangement,
- test case regeneration.

§7.3 Hash Block

$$H_i = (H_{32}^{\text{tag}}, H_{256}^{\text{content}}, H_{562}^{\text{provenance}}). \quad (9)$$

Recommended roles:

Hash	Role
H_{32}^{tag}	Quick human-friendly tag
H_{256}^{content}	Exact record content fingerprint (SHA-256)
$H_{562}^{\text{provenance}}$	Deep lineage signature (custom)

§7.4 Demonstration Table

Object	Seed	Content Hash	Provenance Hash
H element core	fixed canonical	yes	yes
Pu isotope packet	optional	yes	yes
Hexene bead scene	required	yes	yes
Thorium oxalate test cloud	required	yes	yes
Hybrid Pu–Cp–Hexene system	required	yes	yes

§7.5 Demo Entry

Listing 2: Seed and hash block for hybrid scene

```
Object:      hybrid_precursor_scene_004
Seed32:      40128
Seed64:      40128_094_006_005
GeometrySeed: geom_cp_hexene_pu_v2
ChemicalSeed: chem_hybrid_transition_v1
Hash256:     98C1...A442
Hash562:     HYB-PU-CP-HEX-2026-TRACE-....
```

§8 Report 7: Integrated Example Report

§8.1 Goal

This report shows one full input packet combining element identity, material typing, compound registration, precursor staging, and seed and hash traceability.

§8.2 Example: Pu + Cyclopentadienyl + Hexene Hybrid Input

Input packet.

$$I_{\text{demo}} = (E_{\text{Pu}}, C_{\text{Cp-like}}, C_{\text{hexene}}, P_{\text{hybrid}}, S, H). \quad (10)$$

Expanded form.

$$\begin{aligned} E_{\text{Pu}} &= (I, B, N, R, C, X, H), \\ C_{\text{Cp-like}} &= (S, G, B, P, R, H), \\ P_{\text{hybrid}} &= (I, \Psi, \Omega, \Lambda, T, H). \end{aligned}$$

§8.3 Demo Manifest

Listing 3: Integrated demonstration manifest

```
Demo Name: Pu_Cp_Hexene_Hybrid_001
Mode: mixed_resolution
Primary Atomistic Core: Pu
Supporting Precursors: cyclopentadienyl_like, hexene
Environment Class: dry_condensed_transition
Dimensions:
  state_dim = 41
  geom_dim  = 3
  chem_dim  = 19
  nuc_dim   = 5
  aux_dim   = 12
```



```

Data Types:
  core_id      : u32
  runtime_id   : u64
  positions    : f64[3]
  charges      : f64
  oxidation_set : i32[]
  decay_constants : f64[]
  flags        : bool[]
  labels       : str[]
  hash256      : bytes32
  hash562      : custom_trace

Materials:
  plutonium_core
  organic_precursor_family
  organometallic_transition_packet

Compounds:
  hexene
  cyclopentadienyl_like

Precursors:
  Cp_precursor_stage2
  Hexene_bead_stage1

Seed:
  seed32    = 41092
  seed64    = 41092_2026_40C_001
  geom_seed = ring_chain_offset_v1
  chem_seed = hybrid_decay_response_v1

Hash:
  hash256 = ...
  hash562 = ...

```

§9 Report 8: Recommended Chapter Placement

If this material becomes part of a formal textbook or architecture document, the recommended placement is:

Section	Topic
§4.1	Database identity model
§4.2	Input dimensionality and data typing
§4.3	Element, isotope, and material shard layout
§4.4	Compound and precursor registry design
§4.5	Seed and hash provenance
§4.6	Integrated mixed-resolution demonstration packet

§10 Report 9: Clean L^AT_EX Subsection Starter

The following verbatim block is the canonical L^AT_EX subsection starter text for embedding in other documents:

Listing 4: Embeddable L^AT_EX subsection starter

```

\section{Database Architecture Demonstration Reports}

The database layer is treated as a typed scientific registry
rather than a flat storage table. Each input is modeled as
\[
X_i = (I_i, T_i, D_i, P_i, S_i, H_i),
\]
where  $\backslash(I_i\backslash)$  is the identity block,  $\backslash(T_i\backslash)$  is the type block,
 $\backslash(D_i\backslash)$  is the dimension descriptor,  $\backslash(P_i\backslash)$  is the scientific
payload,  $\backslash(S_i\backslash)$  is the deterministic seed block, and  $\backslash(H_i\backslash)$  is
the hash and provenance block.

\subsection{Input Dimension Demonstration}
A dimension descriptor is written as
\[
D_i = (n_{\text{\texttt{state}}}, n_{\text{\texttt{geom}}}, n_{\text{\texttt{chem}}},
      n_{\text{\texttt{nuc}}}, n_{\text{\texttt{aux}}})),
\]
allowing a single registry to host element cores, isotope packets,
atomistic objects, bead objects, materials, compounds, and staged
precursors.

\subsection{Seed and Hash Provenance}
To guarantee reproducibility and traceability, each major object
carries both a deterministic seed block and a multi-scale hash block:
\[
S_i = (\text{\texttt{seed}}_{32}, \text{\texttt{seed}}_{64},
      \text{\texttt{seed}}_{\text{\texttt{geom}}}, \text{\texttt{seed}}_{\text{\texttt{chem}}}),
\]
\[
H_i = (H_{32}^{\text{\texttt{tag}}}, H_{256}^{\text{\texttt{content}}}, H_{562}^{\text{\texttt{provenance}}}).
\]
This allows exact regeneration of synthetic scenes, validation of
imported records, and lineage tracing across atomistic, bead, and
macro-scale projections.

```

§11 Report 10: Best Next Expansions

The next useful reports building on this architecture would be:

1. **Element demo packet** — H, Fe, U, Pu with full $X_i = (I_i, T_i, D_i, P_i, S_i, H_i)$ records, including nuclear descriptor expansions for actinides.
2. **Materials demo packet** — Copper, Steel A36, UO₂, Thorium oxalate with M_k records and engineering property vectors.
3. **Precursor demo packet** — Hexene, cyclopentadienyl, nitrate-metal slurries, dissolved crystal solids, with thermal and strength initial passes for tensile hardness and brittleness if those channels are validated.
4. **Seed reproducibility report** — Demonstrate that the same seed produces the same scene. Show 3D Qt6 rendering with at least two physical representations per atom. Print additional math steps for the reconstruction chain.
5. **Hash mismatch report** — Demonstrate detection of tampered overlays, comparing H_{256}^{content} before and after modification and flagging provenance violations.

Appendix: C++ Header Mapping

The formal definitions in this document are implemented as C++ header-only structures under `coarse_grain/database/`:

Header	Implements
<code>input_record.hpp</code>	X_i , D_i , <code>SemanticType</code> , <code>StorageType</code>
<code>material_record.hpp</code>	M_k
<code>compound_record.hpp</code>	C_j
<code>precursor_record.hpp</code>	P_m
<code>seed_hash.hpp</code>	S_i , H_i
<code>demo_entries.hpp</code>	Canonical demonstration objects